Bihar Engineering University, Patna

B.Tech 6th Semester Examination, 2024

Course : B.Tech
Code : 100607

Subject: Introduction to VLSI Design

Time : 03 Hours
Full Marks : 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer the following (Any seven question only): **[2 x 7 = 14]**

(a) What is the condition for non-saturated region?
(i) Vds = Vgs – Vt
(ii) Vds lesser than Vgs – Vt
(iii) Vgs lesser than Vt
(iv) Vds greater than Vgs – Vt

(b) Inversion layer in enhancement mode consists of excess of _____
(i) positive carriers
(ii) negative carriers
(iii) both in equal quantity
(iv) neutral carriers

(c) What is the design flow of the VLSI system?
i. architecture design ii. market requirement iii. logic design iv. HDL coding
(i) ii-i-iii-iv
(ii) iii-ii-i-iv
(iii) iv-i-iii-ii
(iv) i-ii-iii-iv

(d) If n-transistor conducts and has large voltage between source and drain, then it is said to be in _____ region.
(i) linear
(ii) non saturation
(iii) saturation
(iv) cut-off

(e) Increasing fan-out _____ the propagation delay.
(i) increases
(ii) exponentially decreases
(iii) decreases
(iv) does not affect

(f) Which type of storage element of SRAM is very fast in accessing data but consumes lots of power?
(i) NAND
(ii) NOR
(iii) CMOS
(iv) TTL

(g) In CMOS logic circuit the n-MOS transistor acts as:
(i) Pull down network
(ii) Pull up network
(iii) Load
(iv) Not used in CMOS circuits

(h) The variation of the threshold voltage with the applied bulk-to-source voltage is typically observed by plotting the _____ as a function of the source-to-drain voltage.
(i) drain current
(ii) natural logarithm of the drain current
(iii) square root of the drain current
(iv) square of the drain current

(i) The advantage of BiCMOS circuit is that it gives _____
(i) Small fan-out
(ii) Small fan-in
(iii) Large fan-in
(iv) Large fan-out

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(j) In CMOS logic circuit, the switching operation occurs because:

(i) None of the mentioned

(ii) Both n-MOSFET and p-MOSFET turns ON simultaneously for input '0' and turns OFF simultaneously for input '1'

(iii) Both n-MOSFET and p-MOSFET turns OFF simultaneously for input '0' and turns ON simultaneously for input '1'

(iv) N-MOSFET transistor turns ON, and p-MOSFET transistor turns OFF for input '1' and N-MOS transistor turns OFF, and p-MOS transistor turns ON for input '0'

Q.2 (a) What do you mean by short channel effects in MOSFET? Explain the MOSFET capacitance with suitable sketch. [7]

(b) What do you mean by latch up in CMOS and how it can be prevented? [7]

Q.3 (a) Explain CMOS n-well fabrication process with necessary diagrams. [7]

(b) Explain the DC transfer characteristics of CMOS inverter. Derive the expression of rise time and fall time for CMOS inverter. [7]

Q.4 (a) Explain linear delay model. Compare the logical efforts of the following gates with the help of schematic diagrams: (i) 2-input NAND gate (ii) 3-input NOR gate [7]

(b) Explain the general structure of ratioless synchronous dynamic logic with suitable diagram. [7]

Q.5 (a) Explain pseudo NMOS inverter with suitable diagram. Also write the disadvantages of this circuit. [7]

(b) With necessary circuit diagram, explain the operation of three transistor DRAM cell. [7]

Q.6 (a) With necessary circuit diagram explain 3-bit dynamic shift register with depletion load. [7]

(b) With necessary circuit diagram, explain the operation of tristate inverter. Also realize a 2:1 multiplexer using tristate inverter. [7]

Q.7 (a) Write short note on DRAM cell. Explain leakage and refresh operation in DRAM cells. [7]

(b) Explain BiCMOS logic circuit. Also differentiate between BiCMOS and CMOS circuits. [7]

Q.8 (a) Implement a D flip-flop using transmission gates and explain its operation with necessary timing diagram. [7]

(b) Explain Cascade Voltage Switch Logic(CVSL). Also realize two input AND/NAND using CVSL. [7]

Q.9 (a) Explain the operation of MOSFET with necessary diagrams. Also derive the equation for drain current in linear and saturation region of operation. [7]

(b) Define the following terms: [7]

(i) Channel length modulation

(ii) Parasitic delay

(iii) Noise margin

(iv) Body bias effect in MOSFET

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